

Week 3: Milestone 2

Draft White Paper

Evaluating MLB Trade Deadline Moves Through Predicted WAR

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Business Problem

At the MLB trade deadline there are both a high volume of team trades as well as a high value of quality assets and players that are moved at teams gear up for the playoffs. Some teams look to “sell” current assets in hopes of making their team better in future seasons. Others look to get better now in hopes of winning a championship.

This project seeks to build a framework to evaluate these moves using data driven models to determine true “winners” and “losers” of these trades. This will allow both front offices as well as media outlets to evaluate these trades with data backed models as opposed to arbitrary opinions.

Background/History

The problem lies in how baseball media analyses these trades and then communicates this to baseballs faithful. Media outlets give out letter grades on which team got a better deal out of a trade. Often these grades are based solely on a writer, scout or public opinion of what a player can deliver in terms of value. When statistical analysis is brought up in these evaluations it is often in the terms of basic statistics such as career batting average or how many runs batted in a player has that season. The current analysis does little to project how a player will provide value after the trade and relies heavily on what the player has achieved in the past.

This project will take a true data driven approach to predicting the “winners” and “losers” of trades. This project will project future WAR (wins above replacement) which is the gold standard for how both the media and MLB front offices assess the value a player provides to a team. How many wins that player contributes above a replacement level (minor league) player.

For this project the Scope has been limited to evaluating the pitcher for pitcher trades made by the Minnesota Twins at the 2025 MLB Trade deadline.

Data Explanation

Data for this project was collected from two different sources. The first is pitcher performance data from Statcast, MLB's in-house database for tracking both statistics as well as performance metrics. In this case, all pitcher data has been selected from the 2020 season onwards to train our models for producing future WAR.

After filtering out personally identifiable metrics that may lead the model to overtraining, the dataset contains 207 features ranging from player age to opponent on-base percentage to spin and break metrics for all the pitches in a pitcher's arsenal. Any missing values in the data frame have been replaced with the population median. This allows us to still train on the player's metrics and increases the sample on which we can train on.

The second source for our data comes from FanGraphs, a well-reputable baseball statistics database used by media and front offices alike for their calculation of advanced stats, particularly wins above replacement. Player total WAR was collected from 2020 onwards and joined to the MLB Statcast data on player season and name. From there, the next season WAR was also attached or mapped to each record to train upon it as a target variable.

As seen in Figure 1 below, analyzing the distribution of WAR, we see that the data is skewed right. This is expected as a WAR of zero represents a replacement or minor league player. In the Major Leagues, if a player does not perform at a plus level, they will often be sent back to the minors or released from their contract and thus major league affiliated baseball.

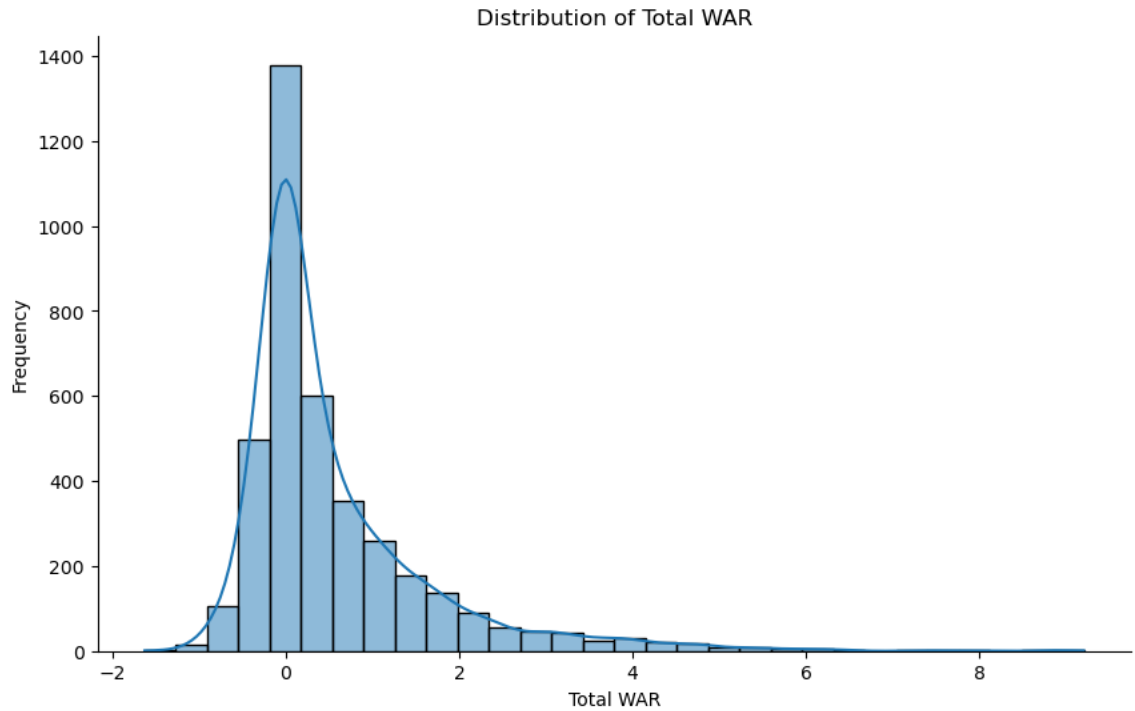


Figure: Distribution of Total WAR MLB 2020-2025

Methods

In order to evaluate on trades on next season projected WAR we first needed to train a model the predicted next season WAR. A combination of backwards stepwise regression (with and without Standard Scaler and PCA). Additionally, a neural network and series of machine learning models were trained through a pipeline using a grid search cross validation to adjust parameters to find the best model across linear, random forest, decision tree, bagging, ada boosting and gradient posting regressions. The best model was a gradient boosting model identifying 10 important features. This model was chosen for evaluation due to its similar performance and less complexity compared to the backwards stepwise regression that included over 120 features as seen in appendix B. All models were trained on an 80-20 train-test split to mitigate and evaluate the effects of overfitting. Evaluating this model on both RMSE and R2

balances the bias variance tradeoff and minimizes the effects of overfitting. Evaluation statistics for the backwards regression model can be seen in appendix a.

The gradient boosting model was likely selected due to its ability to handle the complex data that is the nature of baseball. The feature importance of total WAR predictors can be seen below in figure 3.

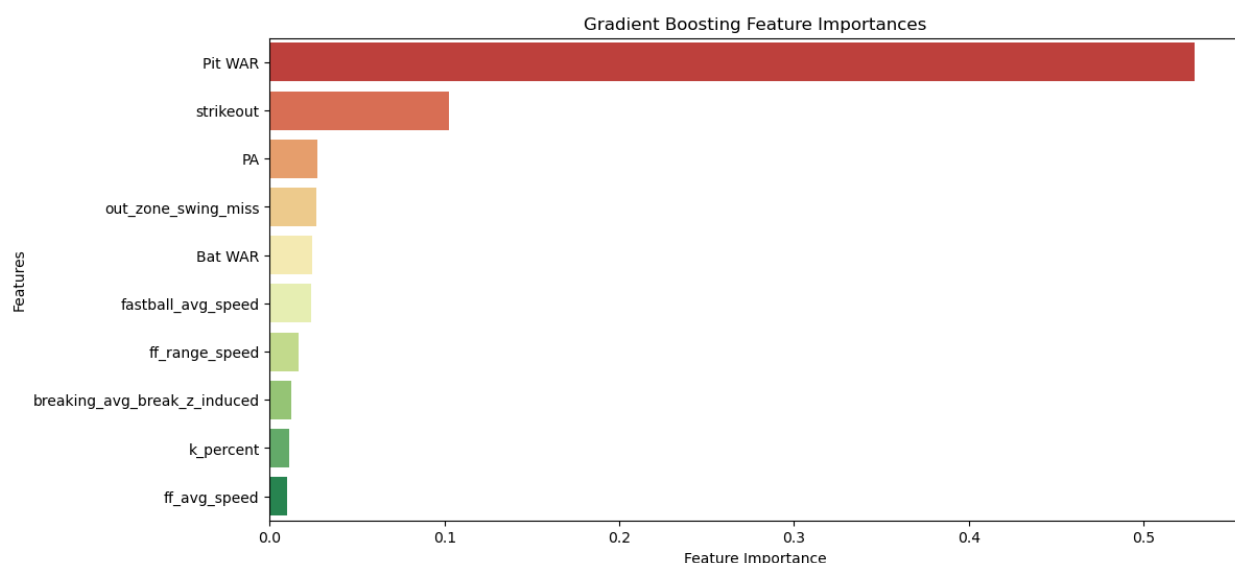


Figure 3: Gradient boosting model feature importance.

Analysis

There were two trades out of 9 made by the Minnesota Twins at the trade deadline that included Major league pitchers on both sides of the trade. Griffin Jax (MIN) for Taj Bradley (TBR) and Jhoan Duran (MIN) for Mick Abel (TOR) and a minor league prospect. The 2025 features were entered into the predictive model, and the projected 2026 WAR was calculated. The results can be seen in figures 4 and 5 below.

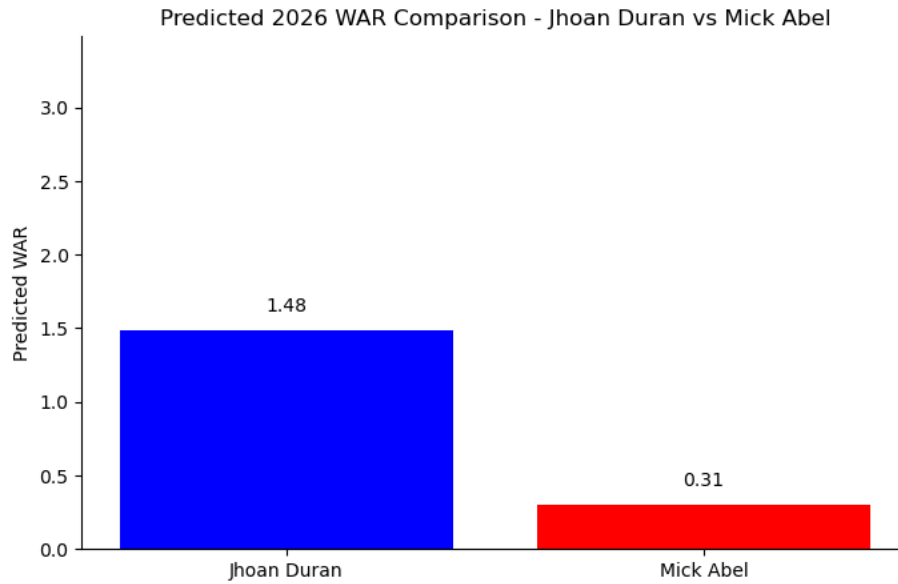


Figure 4: Projected 2026 WAR Jhoan Duran and Mick Abel

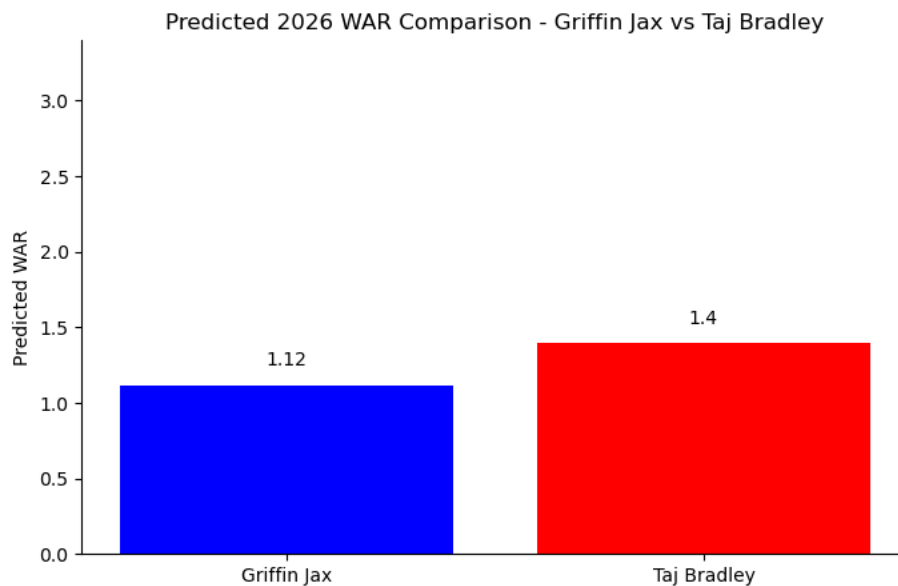


Figure 5: Projected 2026 WAR Griffin Jax and Taj Bradley.

As we can see from the projected 2026 WAR it looks like Minnesota was the clear “loser” of the Duran and “winner” of the Jax trades. This contradicts the common sentiment among writers that Minnesota received poor value in both trades.

Conclusion

When strictly evaluating on next season WAR for determining which team was the winner of a trade. Minnesota is seen as losing the Duran trade and winning the Jax trade. This added context of projected WAR is very helpful for baseball fans as well as front offices as it provides a tangible statistic of how they can evaluate a trade in the near future.

Assumptions

In the training of this model several assumptions were made. One of which being if a player was missing data and the population had that data recorded for 50% of the population the median value was assigned for that player. This assumption has the potential to mislead the model as it helps show the player over or underperforms in an area or gives them performance data for a pitch that they do not throw.

Limitations

The largest limitation of this model is due to the scope of the project and the interest of building one model we can only evaluate trades involving just pitchers. Additionally, this model does not account for the potential value of minor league prospects that are added to deals to provide value. For example, in the Jhoan Duran for Mick Abel trade, Mick Abel was traded along with a minor league prospect catcher. Our model only being trained on MLB stats limits or ability to assign value for minor league players. In this case it may make the trade look more lopsided than it actually is in Torontos favor.

Challenges

Some challenges of this model are that it does a great job of showing some features that may affect or influence future performance but it missis some demographic information such as height and injury history that could make the model more effective and accurate.

Future Uses/Additional Applications

This model can and should be used by both media outlets and the front office. The front office can use it to inform and validate trades before they occur. This can help team ensure they are receiving fair value for their players. For the media this model can help add data backed context for their trade evaluations. This will better inform readers and take some of the ambiguity and emotion out of trade evaluations.

Recommendations

I would recommend that this model be refined to train on minor league stats as well as build a model to predict future WAR for hitters. This will allow the media and front offices to tell more of the story around the trade and properly assign some value to minor league prospects that may prove valuable down the road.

Implementation Plan

I would recommend implementing this model as is in a media application to add context to the trade valuation. I would require the addition of a disclosure that there is no way to assign value for minor league prospects in the trade. I would recommend that front offices implement this tool to help inform trades at the deadline and ensure they are projected to get the value back that they need to part with a player under contract.

Ethical Assessment

There are several ethical considerations when looking at the implementation of this model. The greatest being that this model has the potential to affect players careers. If the model unfairly over or underpredicts their performance, they may be given less opportunity or compensated less. This model implementation also has the potential to affect how teams' front offices are media perceived. If the model incorrectly identity winners or losers team

management and medias social sentiment may be affected for their ability to properly anticipate the perceived value of trades.

Ten Questions an Audience Would Ask.

- 1) What data would you add to your model in attempts to improve accuracy?
 - a. First, I would like to build an additional model that takes minor league player stats and help predict their future MLB WAR so that we can assign a value to the players future value of the team. Additionally, I would like to add in injury history as a feature to help predict if a player's future value may decrease due to injury.
- 2) How would you account for the value of minor league players included in trades?
 - a. I would modify the model to include minor league stats to train on. Additionally, I would have to collect the minor league stats from existing MLB players to train the model on. The data mapping would be difficult but possible.
- 3) Why did you choose to evaluate on WAR and not on another statistic such as ERA or FIP?
 - a. WAR is the most used as well as easily interpreted MLB stat. At one glance you can value how many wins a player is worth whether they are a hitter or pitcher or both. For means of comparison like trade evaluation it is the best option.
- 4) Prospects often don't contribute to major league clubs for several years how would you account for that in future improvements?
 - a. It would be interesting to build a model projecting the future wars for players over the next several seasons and then comparing the totals. This would take a more long-term approach to future value.

5) This model does not include anything about contract values is that important to evaluate and how could it be built in?

- a. A players value is also evaluated in the context of their price. This project being more fan and media facing did not take contract value into account. However, a front office may want to evaluate the trade at a rate of WAR per dollar of paid contract. This would allow teams to find which players are the best value or bargain.

6) Age is not a strong feature in this model does that mean age is not important?

- a. This model is only seeking to predict the value of a player in the immediate next season. Therefore, a players age is likely not a strong feature because aging effects are likely seen over greater periods of time not several months.

7) How does this model value service time before free agency?

- a. This model does not assign any value to a player service time before free agency. Often trades are mad for players that contracts are friendly or “under team control” this model does not evaluate on the value of the contract, just the value the player brings to the on-field performance of the team.

8) If you were a general manager, would you make trades based on this model?

- a. I would not recommend making trades solely on this model because there are shortcomings. For example, the inability to assign value to minor league prospects or the model does not consider any financial repercussions of trades. However, I would use this mode to evaluate prospective trades and ensure that I am receiving enough value in return to be comfortable with the trade.

9) Pitcher war is the strongest importance predictor in your model. Is this overfitting?

- a. No this is not an example of data leaking or overfitting. In fact, it is very realistic for a team or outlet to use a player's previous performance to help predict their future performance. This is the case in this project.

10) Could the implementation of this model have impacts outside the world of baseball?

There is potential for this project to have effects outside of baseball and on the public. For example. If one team is always viewed as being the loser of trades as seen from this model. Over time upset and disinterested fans may stop supporting a team. In extreme cases a team may need to relocate as a result. This would affect the whole economic output of the area as often many other businesses are reliant on traffic and the economic boost a MLB franchise provides to a given market.

Appendix: Supporting Documentation

Appendix A: Performance stats gradient boosting model.

```
Best Estimator Train R²: 0.5554431361687824
Best Estimator Train RMSE: 0.7883023952485619
Best Estimator Test R²: 0.3460816575047302
Best Estimator Test RMSE: 0.9472162006588407
```

Appendix B: Backwards stepwise regression model summary:

Final model summary:

OLS Regression Results

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Dep. Variable:  next_season_total_war  R-squared:          0.497
Model:          OLS  Adj. R-squared:      0.461
Method:         Least Squares  F-statistic:      13.88
Date:           Sun, 28 Sep 2025  Prob (F-statistic):  8.22e-177
Time:           15:16:08  Log-Likelihood:    -2247.4
No. Observations: 1808  AIC:              4737.
Df Residuals:     1687  BIC:              5402.
Df Model:          120
Covariance Type:  nonrobust
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```

	coef	std err	t	P> t	[0.025	0.975]
const	-84.6367	257.946	-0.328	0.743	-590.564	421.291
year	-0.0237	0.023	-1.039	0.299	-0.068	0.021
player_age	-0.0091	0.007	-1.313	0.189	-0.023	0.004
p_game	-0.0069	0.002	-3.871	0.000	-0.010	-0.003
p_formatted_ip	-0.0199	0.019	-1.072	0.284	-0.056	0.017
pa	0.0241	0.008	2.846	0.004	0.008	0.041
ab	-0.0223	0.008	-2.626	0.009	-0.039	-0.006
single	0.0142	0.007	2.087	0.037	0.001	0.028
double	-0.0133	0.009	-1.467	0.143	-0.031	0.004
triple	0.0150	0.022	0.693	0.488	-0.028	0.058
strikeout	0.0174	0.003	5.662	0.000	0.011	0.023
walk	-0.0247	0.009	-2.720	0.007	-0.043	-0.007
on_base_plus_slg	-13.0223	14.556	-0.895	0.371	-41.571	15.527
babip	3.8082	3.013	1.264	0.206	-2.100	9.717
p_total_stolen_base	0.0054	0.006	0.874	0.382	-0.007	0.018
p_lob	-0.0079	0.003	-2.372	0.018	-0.014	-0.001
xba	15.5728	15.079	1.033	0.302	-14.002	45.148
xwoba	-5.4252	5.310	-1.022	0.307	-15.839	4.989
xobp	16.5819	15.098	1.098	0.272	-13.030	46.194
xiso	12.8352	14.547	0.882	0.378	-15.697	41.367
wobacon	-10.3442	7.972	-1.298	0.195	-25.980	5.291
xwobacon	15.0930	8.070	1.870	0.062	-0.735	30.921

bacon	9.6890	10.277	0.943	0.346	-10.468	29.846
xbacon	-18.2628	9.292	-1.966	0.050	-36.487	-0.038
xbadiff	-7.8854	9.019	-0.874	0.382	-25.574	9.804
xslgdiff	15.6371	12.423	1.259	0.208	-8.729	40.003
wobadiff	12.2463	11.805	1.037	0.300	-10.908	35.401
squared_up_contact	0.0123	0.014	0.847	0.397	-0.016	0.041
squared_up_swing	-0.0182	0.017	-1.079	0.281	-0.051	0.015
avg_swing_length	-0.1487	0.188	-0.791	0.429	-0.518	0.220
swords	-0.0088	0.006	-1.490	0.136	-0.020	0.003
sweet_spot_percent	-0.0092	0.007	-1.378	0.169	-0.022	0.004
barrel_batted_rate	0.0896	0.034	2.652	0.008	0.023	0.156
solidcontact_percent	0.0731	0.033	2.244	0.025	0.009	0.137
flareburner_percent	0.0620	0.032	1.964	0.050	8.28e-05	0.124
poorlyunder_percent	0.0610	0.032	1.900	0.058	-0.002	0.124
poorlytopped_percent	0.0602	0.031	1.919	0.055	-0.001	0.122
poorlyweak_percent	0.0564	0.032	1.753	0.080	-0.007	0.119
avg_best_speed	-0.0112	0.014	-0.802	0.423	-0.038	0.016
avg_hyper_speed	-0.0329	0.043	-0.765	0.444	-0.117	0.051
edge_percent	0.0094	0.008	1.183	0.237	-0.006	0.025
f_strike_percent	0.0091	0.004	2.269	0.023	0.001	0.017
groundballs_percent	0.2591	0.222	1.168	0.243	-0.176	0.694
flyballs_percent	0.2560	0.222	1.153	0.249	-0.179	0.691
linedrives_percent	0.2554	0.222	1.151	0.250	-0.180	0.691
popups_percent	0.2514	0.222	1.132	0.258	-0.184	0.687
n_ff_formatted	-0.0035	0.002	-1.526	0.127	-0.008	0.001
ff_avg_speed	0.0481	0.021	2.334	0.020	0.008	0.088
ff_avg_break_z	-0.0182	0.012	-1.546	0.122	-0.041	0.005
sl_avg_speed	-0.1608	0.115	-1.392	0.164	-0.387	0.066
sl_avg_spin	-0.0002	0.000	-0.801	0.423	-0.001	0.000
sl_avg_break_x	-0.0064	0.006	-1.149	0.251	-0.017	0.005
sl_avg_break_z	0.1734	0.116	1.494	0.135	-0.054	0.401
sl_avg_break_z_induced	-0.1792	0.116	-1.548	0.122	-0.406	0.048
n_ch_formatted	-0.0259	0.014	-1.789	0.074	-0.054	0.003
ch_avg_speed	-0.6009	0.218	-2.751	0.006	-1.029	-0.173
ch_avg_break_x	-0.0246	0.017	-1.473	0.141	-0.057	0.008
ch_avg_break_z	0.7016	0.255	2.749	0.006	0.201	1.202
ch_avg_break_z_induced	-0.7176	0.251	-2.856	0.004	-1.210	-0.225
ch_avg_break	0.0466	0.030	1.576	0.115	-0.011	0.105
n_cu_formatted	-0.0024	0.003	-0.714	0.476	-0.009	0.004
cu_avg_speed	-0.1190	0.076	-1.558	0.119	-0.269	0.031
cu_avg_break_x	-0.0056	0.004	-1.273	0.203	-0.014	0.003
cu_avg_break_z	0.1178	0.067	1.758	0.079	-0.014	0.249
cu_avg_break_z_induced	-0.1263	0.066	-1.906	0.057	-0.256	0.004
n_si_formatted	-0.0029	0.003	-1.125	0.261	-0.008	0.002
si_avg_speed	0.2258	0.108	2.091	0.037	0.014	0.438
si_avg_break_x	-0.0043	0.004	-1.157	0.247	-0.012	0.003
si_avg_break_z	-0.3357	0.155	-2.162	0.031	-0.640	-0.031
si_avg_break_z_induced	0.3344	0.157	2.128	0.033	0.026	0.643
si_avg_break	-0.0295	0.020	-1.483	0.138	-0.068	0.009
n_fc_formatted	-0.0081	0.003	-2.758	0.006	-0.014	-0.002
fc_avg_speed	0.0289	0.014	2.004	0.045	0.001	0.057
fc_range_speed	-0.1456	0.145	-1.007	0.314	-0.429	0.138
n_fs_formatted	-0.0234	0.014	-1.721	0.085	-0.050	0.003
fs_avg_speed	-0.2001	0.184	-1.085	0.278	-0.562	0.162
fs_avg_spin	0.0003	0.000	1.315	0.189	-0.000	0.001

fs_avg_break_x	-0.0267	0.014	-1.843	0.065	-0.055	0.002
fs_avg_break_z	0.1944	0.211	0.921	0.357	-0.220	0.609
fs_avg_break_z_induced	-0.1875	0.212	-0.885	0.376	-0.603	0.228
n_kn_formatted	-0.5906	0.312	-1.895	0.058	-1.202	0.021
kn_avg_speed	3.1048	1.810	1.715	0.086	-0.446	6.655
kn_avg_spin	0.0070	0.006	1.162	0.245	-0.005	0.019
kn_avg_break_z	-2.5727	1.409	-1.826	0.068	-5.336	0.191
kn_avg_break_z_induced	2.1055	1.390	1.515	0.130	-0.621	4.832
kn_avg_break	1.1296	0.628	1.799	0.072	-0.102	2.361
kn_range_speed	11.6489	8.379	1.390	0.165	-4.785	28.082
st_avg_speed	-0.1824	0.117	-1.556	0.120	-0.412	0.048
st_avg_spin	-0.0006	0.000	-2.499	0.013	-0.001	-0.000
st_avg_break_x	-0.0027	0.004	-0.751	0.453	-0.010	0.004
st_avg_break_z	0.2283	0.109	2.101	0.036	0.015	0.441
st_avg_break_z_induced	-0.2353	0.109	-2.167	0.030	-0.448	-0.022
st_avg_break	0.0328	0.018	1.861	0.063	-0.002	0.067
st_range_speed	-0.2468	0.179	-1.382	0.167	-0.597	0.104
sv_avg_spin	-0.0008	0.000	-1.713	0.087	-0.002	0.000
sv_avg_break	0.0425	0.040	1.052	0.293	-0.037	0.122
sv_range_speed	-0.5149	0.678	-0.759	0.448	-1.845	0.816
n_fo_formatted	0.1763	0.145	1.218	0.223	-0.108	0.460
fo_avg_speed	-2.8002	1.290	-2.171	0.030	-5.330	-0.270
fo_avg_spin	0.0166	0.008	2.066	0.039	0.001	0.032
fo_avg_break_x	2.5024	1.567	1.597	0.110	-0.571	5.576
n_sc_formatted	-3.4298	10.229	-0.335	0.737	-23.493	16.634
sc_avg_break_z	0.5537	1.311	0.422	0.673	-2.017	3.125
sc_range_speed	-1.7149	5.115	-0.335	0.737	-11.747	8.317
n_fastball_formatted	0.0070	0.003	2.457	0.014	0.001	0.013
fastball_avg_spin	0.0003	0.000	1.482	0.139	-9.3e-05	0.001
fastball_avg_break_x	0.0089	0.005	1.724	0.085	-0.001	0.019
fastball_avg_break	0.0262	0.016	1.644	0.100	-0.005	0.057
breaking_avg_speed	0.2050	0.107	1.909	0.056	-0.006	0.416
breaking_avg_spin	0.0002	0.000	1.019	0.308	-0.000	0.001
breaking_avg_break_z	-0.2192	0.102	-2.158	0.031	-0.418	-0.020
breaking_avg_break_z_induced	0.2079	0.102	2.041	0.041	0.008	0.408
breaking_range_speed	0.0734	0.028	2.653	0.008	0.019	0.128
n_offspeed_formatted	0.0322	0.014	2.224	0.026	0.004	0.061
offspeed_avg_speed	0.5699	0.209	2.725	0.007	0.160	0.980
offspeed_avg_break_x	0.0235	0.017	1.374	0.170	-0.010	0.057
offspeed_avg_break_z	-0.6560	0.245	-2.678	0.007	-1.137	-0.176
offspeed_avg_break_z_induced	0.6631	0.241	2.749	0.006	0.190	1.136
offspeed_avg_break	-0.0303	0.028	-1.090	0.276	-0.085	0.024
offspeed_range_speed	0.2958	0.099	2.972	0.003	0.101	0.491
PA	0.0041	0.002	2.060	0.040	0.000	0.008
Bat WAR	0.5157	0.266	1.938	0.053	-0.006	1.038
Pit WAR	0.2570	0.064	4.034	0.000	0.132	0.382

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Omnibus:	270.603	Durbin-Watson:	2.004
Prob(Omnibus):	0.000	Jarque-Bera (JB):	814.177
Skew:	0.765	Prob(JB):	1.60e-177
Kurtosis:	5.909	Cond. No.	1.00e+16

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Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The smallest eigenvalue is $6.77e-22$. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.

Appendix C: Sources

Baseball savant: Statcast, trending MLB players and visualizations. baseballsavant.com. (n.d.).
<https://baseballsavant.mlb.com/>

Matthew Taylor has been a writer with Twins Daily since 2020. If you can't find him at Target Field you'll most likely find him at the movie theater. Follow Matthew Taylor on Twitter: MatthewTaylorMN. (2025, August 11). *Grading the 9 trades from the Twins' 2025 trade deadline.* Twins Daily. <https://twinsdaily.com/news-rumors/minnesota-twins/grading-the-9-trades-from-the-twins-2025-trade-deadline-r18916/>

War leaderboards. FanGraphs Baseball. (n.d.).
<https://www.fangraphs.com/leaders/war?season=2024>